

Maximum Feasible Recycled Content and Scenario Effects

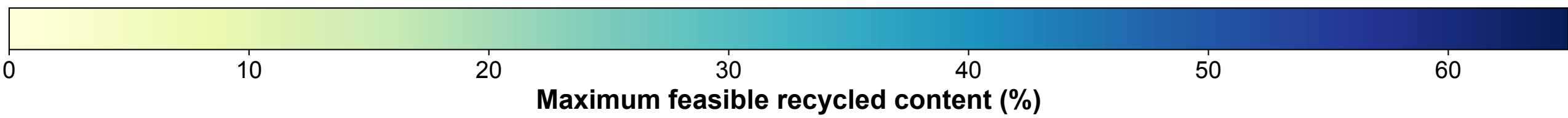
a) Reference values

Baseline Policy · Increase LIB Cap · Benchmark Chemistry

California CA-origin feedstock / US recycling network					
Ni					
	1.9	5.7	17.9	34.3	56.6
	2.0	4.5	15.1	34.5	56.8
	2.5	6.8	21.3	39.6	61.2
Li	1.5	4.6	14.6	30.0	48.1
Gr	1.6	5.0	16.1	33.4	53.7
Cu	1.6	5.2	16.9	35.5	56.9
2030 2035 2040 2045 2050					

United States US-origin feedstock / US recycling network					
	1.2	3.7	13.0	28.9	50.2
	1.2	3.1	11.8	29.3	50.4
	1.4	4.2	15.4	33.0	54.0
	1.1	3.5	11.9	26.4	43.8
	1.2	4.0	13.2	29.7	49.2
	1.3	4.3	14.2	31.9	52.3
2030 2035 2040 2045 2050					

North America NA-origin feedstock / NA recycling network					
	1.1	3.7	13.5	31.1	55.0
	1.2	3.1	12.3	31.5	55.2
	1.4	4.2	16.1	35.6	59.3
	1.0	3.5	12.1	27.8	46.8
	1.1	3.9	13.5	31.0	52.3
	1.2	4.2	14.4	33.2	55.4
2030 2035 2040 2045 2050					



b) Scenario changes

ΔPolicy = Rollback – Baseline | ΔCapacity = Decrease – Increase | ΔChemistry = High-LFP – Benchmark | Cell: Δ change (Reference + Δ)

California CA-origin feedstock / US recycling network						
Ni	ΔPolicy	+1.8 (3.7)	+3.3 (9.0)	+4.3 (22.1)	-4.3 (30.0)	-6.9 (49.7)
	ΔCapacity	+0.7 (2.6)	+3.6 (9.3)	+10.8 (28.6)	+10.9 (45.2)	+6.2 (62.8)
	ΔChemistry	+0.4 (2.3)	+1.9 (7.6)	+7.5 (25.3)	+8.9 (43.2)	+7.1 (63.7)
Mn	ΔPolicy	+1.9 (3.8)	+2.4 (6.9)	+2.7 (17.8)	-4.7 (29.8)	-6.9 (49.9)
	ΔCapacity	+0.7 (2.6)	+2.7 (7.2)	+8.6 (23.7)	+10.7 (45.3)	+6.2 (63.0)
	ΔChemistry	+0.4 (2.3)	+1.5 (5.9)	+6.1 (21.2)	+8.8 (43.3)	+7.1 (63.9)
Co	ΔPolicy	+2.5 (5.0)	+4.0 (10.8)	+4.9 (26.1)	-5.4 (34.2)	-7.8 (53.5)
	ΔCapacity	+0.9 (3.4)	+4.4 (11.2)	+13.1 (34.4)	+13.0 (52.7)	+6.9 (68.1)
	ΔChemistry	+0.5 (3.0)	+2.3 (9.1)	+9.0 (30.3)	+10.5 (50.2)	+7.9 (69.1)
Li	ΔPolicy	+1.2 (2.7)	+2.3 (6.9)	+3.6 (18.3)	-2.0 (28.0)	-4.1 (44.0)
	ΔCapacity	+0.4 (1.9)	+2.3 (6.8)	+6.5 (21.1)	+7.6 (37.6)	+4.5 (52.6)
	ΔChemistry	+0.0 (1.5)	+0.1 (4.7)	+0.3 (14.9)	+0.4 (30.3)	+0.2 (48.3)
Gr	ΔPolicy	+1.2 (2.8)	+2.5 (7.5)	+4.0 (20.1)	-1.9 (31.4)	-4.3 (49.4)
	ΔCapacity	+0.5 (2.1)	+2.4 (7.4)	+6.9 (23.0)	+8.2 (41.6)	+5.0 (58.7)
	ΔChemistry	-0.0 (1.6)	-0.1 (4.9)	-0.2 (15.9)	-0.2 (33.1)	-0.2 (53.6)
Cu	ΔPolicy	+1.2 (2.8)	+2.5 (7.7)	+4.3 (21.1)	-1.6 (34.0)	-4.1 (52.8)
	ΔCapacity	+0.5 (2.1)	+2.3 (7.6)	+6.7 (23.6)	+8.2 (43.8)	+5.1 (61.9)
	ΔChemistry	-0.1 (1.6)	-0.3 (4.9)	-1.1 (15.8)	-1.5 (34.0)	-1.2 (55.7)
		2030	2035	2040	2045	2050

United States US-origin feedstock / US recycling network					
	+0.3 (1.5)	+0.6 (4.2)	+0.3 (13.3)	-2.3 (26.5)	-3.0 (47.2)
	+0.4 (1.6)	+2.2 (5.9)	+7.4 (20.4)	+8.9 (37.7)	+5.5 (55.6)
	+0.2 (1.4)	+1.2 (4.8)	+5.1 (18.1)	+7.2 (36.0)	+6.2 (56.3)
	+0.4 (1.6)	+0.4 (3.5)	+0.1 (11.9)	-2.4 (26.9)	-3.0 (47.4)
	+0.4 (1.7)	+1.8 (4.9)	+6.3 (18.1)	+8.8 (38.1)	+5.5 (55.9)
	+0.2 (1.5)	+0.9 (4.0)	+4.5 (16.2)	+7.2 (36.4)	+6.2 (56.6)
	+0.4 (1.9)	+0.7 (4.8)	+0.2 (15.6)	-2.8 (30.2)	-3.3 (50.7)
	+0.5 (1.9)	+2.6 (6.8)	+9.0 (24.4)	+10.4 (43.4)	+5.9 (59.9)
	+0.3 (1.7)	+1.4 (5.5)	+6.2 (21.6)	+8.4 (41.4)	+6.8 (60.8)
	+0.2 (1.3)	+0.5 (4.1)	+1.0 (12.9)	-0.8 (25.7)	-1.6 (42.2)
	+0.3 (1.4)	+1.5 (5.0)	+4.6 (16.5)	+6.3 (32.7)	+4.3 (48.1)
	+0.0 (1.1)	+0.1 (3.6)	+0.2 (12.1)	+0.3 (26.7)	+0.2 (44.1)
	+0.2 (1.4)	+0.6 (4.6)	+1.2 (14.5)	-0.7 (29.0)	-1.7 (47.5)
	+0.3 (1.5)	+1.6 (5.6)	+4.9 (18.2)	+6.8 (36.5)	+4.7 (53.9)
	-0.0 (1.2)	-0.0 (3.9)	-0.1 (13.1)	-0.2 (29.5)	-0.2 (49.0)
	+0.2 (1.5)	+0.7 (5.0)	+1.5 (15.8)	-0.3 (31.5)	-1.5 (50.9)
	+0.3 (1.6)	+1.6 (6.0)	+4.8 (19.1)	+6.8 (38.7)	+4.9 (57.2)
	-0.0 (1.2)	-0.2 (4.1)	-0.8 (13.4)	-1.3 (30.6)	-1.1 (51.2)
2030 2035 2040 2045 2050					

North America NA-origin feedstock / NA recycling network					
	+0.4 (1.5)	+0.7 (4.3)	+0.4 (13.9)	-2.3 (28.8)	-3.2 (51.8)
	+0.4 (1.5)	+2.2 (5.8)	+7.5 (21.1)	+9.3 (40.3)	+5.6 (60.6)
	+0.2 (1.3)	+1.1 (4.8)	+5.2 (18.7)	+7.5 (38.6)	+6.5 (61.5)
	+0.4 (1.6)	+0.5 (3.6)	+0.2 (12.5)	-2.3 (29.2)	-3.2 (52.0)
	+0.4 (1.6)	+1.8 (4.9)	+6.5 (18.8)	+9.2 (40.7)	+5.6 (60.8)
	+0.2 (1.4)	+0.9 (4.0)	+4.6 (16.9)	+7.5 (39.0)	+6.5 (61.7)
	+0.5 (1.9)	+0.8 (5.0)	+0.3 (16.4)	-2.7 (32.8)	-3.6 (55.7)
	+0.5 (1.9)	+2.6 (6.8)	+9.2 (25.2)	+11.0 (46.5)	+6.2 (65.5)
	+0.3 (1.6)	+1.4 (5.6)	+6.3 (22.4)	+8.9 (44.4)	+7.2 (66.5)
	+0.3 (1.3)	+0.6 (4.1)	+1.1 (13.2)	-0.7 (27.0)	-1.5 (45.3)
	+0.3 (1.3)	+1.5 (5.0)	+4.6 (16.7)	+6.2 (33.9)	+3.7 (50.5)
	+0.0 (1.0)	+0.1 (3.5)	+0.2 (12.3)	+0.3 (28.0)	+0.2 (47.0)
	+0.3 (1.4)	+0.7 (4.6)	+1.3 (14.8)	-0.6 (30.4)	-1.6 (50.8)
	+0.3 (1.4)	+1.6 (5.5)	+4.8 (18.3)	+6.7 (37.7)	+4.1 (56.4)
	-0.0 (1.1)	-0.0 (3.9)	-0.1 (13.3)	-0.2 (30.8)	-0.2 (52.2)
	+0.3 (1.5)	+0.8 (5.0)	+1.6 (16.0)	-0.3 (32.9)	-1.3 (54.1)
	+0.3 (1.5)	+1.6 (5.8)	+4.7 (19.2)	+6.6 (39.8)	+4.1 (59.5)
	-0.0 (1.2)	-0.2 (4.0)	-0.8 (13.6)	-1.3 (31.9)	-1.0 (54.4)
2030 2035 2040 2045 2050					

